CYBERSRS

Secure Monitoring Platform Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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Contents

1. Introduction 3

2. Functional requirements 4

3. Non-functional requirements 5

4. Security, privacy and compliance 6

5. Data requirements 6

6. Network and interfaces 6

7. Compliance and policy applicability matrix 6

8. Architecture 7

9. Threats 8

10. Risks 9

11. User stories and use cases 9

12. Verification and acceptance 9

13. Glossary 10

14. SDLC recommendation and workflow 11

SDLC rankings and decision evidence 13

Evidence register 14

Source references 15

Validation findings 16

Human review history 16

AI drafting record 16

Requirements traceability matrix 18

1. Introduction

Original project description

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments. The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes. Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary. The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds.

Statement: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A university server with active monitoring enabled

When: System events are generated and transmitted to the platform

Then: All event types are received and stored in a normalized format

Method: test

Observable evidence: Test capture of event ingestion from a simulated server

Unresolved decisions: Data feed encryption method; Event normalization schema

FR-002 - detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.

Statement: The system shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A user attempting multiple failed logins on a managed endpoint

When: The platform processes authentication attempts and compares them to baseline behavior

Then: An alert is generated for repeated failed authentication events

Method: demonstration

Observable evidence: Test capture of alert generation during simulated attack

Unresolved decisions: Threshold configuration values; Baseline data collection frequency

FR-003 - Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Statement: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator with access privileges and a test alert from a managed device

When: The administrator requests alert review and performs an investigation

Then: The system returns the alert details and allows status update and endpoint isolation

Method: test

Observable evidence: Test capture of alert response and status change

Unresolved decisions: Endpoint isolation workflow; Access control for investigation actions

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Statement: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A security administrator accesses the system with valid credentials

When: The administrator requests to view audit logs and security status dashboards

Then: The system returns audit logs of security events and administrator actions, and displays a centralized dashboard showing current security status across multiple campus locations

Method: test

Observable evidence: Test capture of the dashboard display and log retrieval response

Unresolved decisions: The specific data fields included in audit logs; The granularity of security status indicators on the dashboard; Whether real-time updates are provided to the dashboard

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-003

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-003; NFR-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

7. Compliance and policy applicability matrix

Open decisions: Which specific data fields should be included in audit logs?; What is the granularity of security status indicators on the centralized dashboard?; Should real-time updates be provided to the dashboard?; How will endpoint isolation workflows be defined and executed?; What access controls will govern investigation actions during incident response?; What encryption method will be used for data feeds between departments and servers?; What schema will be used to normalize collected events from multiple sources?; What threshold configuration values will trigger alerts for unusual behavior?; What baseline data collection frequency will be maintained for behavioral analysis?

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Security administrators

Detail: Proposed logical endpoint for the source-stated actor: Security administrators.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Component name: Actor: Security administrators

Responsibility: Represent the source-stated actor: Security administrators.

Event Collection Engine

Detail: Proposed logical owner for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-001; NFR-001

Component name: Event Collection Engine

Responsibility: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds. The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Anomaly Detection Module

Detail: Proposed logical owner for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Component name: Anomaly Detection Module

Responsibility: The system shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.

Security Investigation Workflow

Detail: Proposed logical owner for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Component name: Security Investigation Workflow

Responsibility: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

From component: Actor: Security administrators

To component: Security Investigation Workflow

Data or action: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Untrusted component: Actor: Security administrators

Trusted component: Security Investigation Workflow

Enforcement: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A university server with active monitoring enabled
2. When: System events are generated and transmitted to the platform

Expected result: Expected: All event types are received and stored in a normalized format

Evidence to record: Evidence: Test capture of event ingestion from a simulated server

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A user attempting multiple failed logins on a managed endpoint
2. When: The platform processes authentication attempts and compares them to baseline behavior

Expected result: Expected: An alert is generated for repeated failed authentication events

Evidence to record: Evidence: Test capture of alert generation during simulated attack

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: An administrator with access privileges and a test alert from a managed device
2. When: The administrator requests alert review and performs an investigation

Expected result: Expected: The system returns the alert details and allows status update and endpoint isolation

Evidence to record: Evidence: Test capture of alert response and status change

Test type: positive

Id: TEST-004

Requirement ids: NFR-001

Procedure

1. Given: A security administrator accesses the system with valid credentials
2. When: The administrator requests to view audit logs and security status dashboards

Expected result: Expected: The system returns audit logs of security events and administrator actions, and displays a centralized dashboard showing current security status across multiple campus locations

Evidence to record: Evidence: Test capture of the dashboard display and log retrieval response

Test type: positive

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to favor one SDLC model over others. All models receive equal suitability points (40) due to absence of factors such as budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. Unknown factors prevent a definitive selection; the decision remains provisional until additional evidence is available.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds.; FR-002: The system shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.; FR-003: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.
2. Review and baseline the source-defined work group: NFR-001: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments via secure data feeds.; FR-002: The system shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes using predefined behavioral thresholds.; FR-003: Security administrators shall be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.
3. Implement the source-defined work group: NFR-001: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.
4. In an authorized isolated test environment, execute TEST-001: Given: A university server with active monitoring enabled When: System events are generated and transmitted to the platform Expected: All event types are received and stored in a normalized format
5. In an authorized isolated test environment, execute TEST-002: Given: A user attempting multiple failed logins on a managed endpoint When: The platform processes authentication attempts and compares them to baseline behavior Expected: An alert is generated for repeated failed authentication events
6. In an authorized isolated test environment, execute TEST-003: Given: An administrator with access privileges and a test alert from a managed device When: The administrator requests alert review and performs an investigation Expected: The system returns the alert details and allows status update and endpoint isolation
7. In an authorized isolated test environment, execute TEST-004: Given: A security administrator accesses the system with valid credentials When: The administrator requests to view audit logs and security status dashboards Expected: The system returns audit logs of security events and administrator actions, and displays a centralized dashboard showing current security status across multiple campus locations
8. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
9. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Id: E-002

Source: description

Quote: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Id: E-003

Source: description

Quote: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Id: E-004

Source: description

Quote: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7468

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7429

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7322

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7183

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.7274

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:19 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 19

Page or section: text chunk 20

Relevance score: 0.7272

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.29

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.29